

LSC Kingwood Fire Science — Water Line & Septic System

CSP 26-08-10 | Materials/quantity breakdown for human estimator review | Report-only, no bid submitted

Report-only. This is a materials/quantity breakdown generated from the real bid attachments for internal review. It does not submit a bid, price a proposal, or commit ENGCO to anything — every line is either cited to an exact source or explicitly flagged as needing human takeoff.

21 total line items	18 cited/verified	3 flagged for human takeoff
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Div	Item	Qty	Unit	Confidence	Source
33	Kingwood Drive segment, 12-inch C900 PVC (STA 1+00.00 to STA 6+00.00, tie-in to wet connection) No bid-quantity sheet exists in this set and no single station is printed as "end of line," but the sheet's own survey curve data gives an exact length by arithmetic, not a guess: 59.83 ft (STA 1+00 to PC) + 299.25 ft (curve arc, PC to PT) + 140.89 ft (tangent past PT, per the printed bearing+distance call) = 499.97 LF. Cross-checked a second way, by straight station subtraction (end STA 6+00.00 [=PT 4+59.07 + 140.89] minus begin STA 1+00.00) = 499.96 LF. The two independent computations agree to within 0.01 ft — real closure, not rounding luck. Also visually confirmed against the rendered plan sheet (not just trusted from text): the 'PROP 12" PVC (C-900) WATER LINE (PUBLIC)' leader points directly at the dashed line that sweeps through this exact curve, and that same line runs unbroken through the tangent to the '1 - 12" WET CONNECTION' callout at the far end — the hydrant labeled 'STA 4+59' sits exactly where the curve visually straightens, matching the printed PT: 4+59.07. This whole segment is 12-inch pipe only; no 8-inch is called out on this sheet (Sorters Rd is where the 12-to-8 transition happens, on the next line).	499.96	lf	cross_checked	26 0813 - LSC Kingwood FTF_Aproved PUB Plans_260813.pdf p4 — STA. 1+00.00 KINGWOOD DRIVE (baseline origin); PC: 1+59.83; L=299.25 R=2150.000 T=149.865 (curve data); PT: 4+59.07; tangent-out bearing N88°13'09.99"E, distance 140.89 (all printed directly on the plan-and-profile sheet)
33	Sorters Road segment, 12-inch/8-inch C900 PVC (STA 1+00.00 to STA 4+13.47, tie-in to trenchless crossing) 313.47 LF = 413.47 - 100.00, station arithmetic between two independently, explicitly printed station callouts (both sheets agree on the tie-in station exactly). This is the one segment length on this project defensible enough to call cross_checked rather than a plan-scaled estimate. Includes the 12-inch-to-8-inch transition; sheet 05's profile shows the transition graphically but does not print its exact station.	313.47	lf	cross_checked	26 0813 - LSC Kingwood FTF_Aproved PUB Plans_260813.pdf p5 — STA. 1+00.00 SORTERS ROAD (baseline origin, shared with Kingwood Drive STA 1+00.00) 26 0813 - LSC Kingwood FTF_Aproved PUB Plans_260813.pdf p6 — STA. 4+13.47 SORTERS ROAD = STA. 1+34.68 SORTERS ROAD CROSSING (explicit tie-in station, printed on both sheet 05 and sheet 06)
33	Trenchless 8-inch C900 PVC crossing under Sorters Road Real independent agreement: the SOW's 'approximately 44 linear feet' and the plan's printed '44 LF' label match exactly.	44	lf	cross_checked	26 0814 SOW Water Line 12 & 8 RV1.pdf p4 — Construct approximately 44 linear feet of trenchless 8-inch C900 PVC public water line crossing Sorters Road... 26 0813 - LSC Kingwood FTF_Aproved PUB Plans_260813.pdf p6 — PROP. 44 LF TRENCHLESS CONSTRUCTION (PUBLIC) — printed directly on the Sorters Road Crossing plan-and-profile sheet

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33	Ductile iron pipe transition sections (AWWA C151, C104 lining, 8-mil poly wrap) Real requirement, shown graphically (hatched) on sheets 04-06 profiles at each vertical-bend/casing location, but no length is labeled at any transition. Needs a human take-off from the profile details.	—	lf	unresolved	26 0814 SOW Water Line 12 & 8 RV1.pdf p4 — <i>Provide required transitions between PVC and ductile iron pipe and fittings...</i> 26 0813 - LSC Kingwood FTF_Aproved PUB Plans_260813.pdf p4 — <i>1 - 12" WET CONNECTION / REMOVE EXIST. PLUG AND BOV AND CONNECT PROP. 12" WATER LINE TO EXIST. 12" WATER LINE — printed callout, east end of Kingwood Drive sheet</i>
33	12-inch wet connection to existing City water line	1	ea	cross_checked	26 0814 SOW Water Line 12 & 8 RV1.pdf p4 — <i>Perform one 12-inch wet connection to the existing 12-inch City water line on Kingwood Drive...</i> 26 0813 - LSC Kingwood FTF_Aproved PUB Plans_260813.pdf p4 — <i>1 - 12" WET CONNECTION / REMOVE EXIST. PLUG AND BOV AND CONNECT PROP. 12" WATER LINE TO EXIST. 12" WATER LINE — printed callout, east end of Kingwood Drive sheet</i>
33	Fire hydrant assemblies (6-inch lead, auxiliary valve, thrust restraint) Corrected from an earlier vision-only count of 4, which was wrong. Re-derived properly: text-searched every sheet for the real 'F.H. (STA X+XX)' station callouts (3 found: 1+44, 4+59, 4+18), then cross-checked against the profile views' own 'PROP. 6" F.H. LEAD' callouts (2 on sheet 04, 1 on sheet 05 = 3, matching exactly). Two independent counts agreeing at 3, not 4 — this is why a vision-only count was flagged for re-confirmation instead of trusted the first time.	3	ea	cross_checked	26 0813 - LSC Kingwood FTF_Aproved PUB Plans_260813.pdf p4 — <i>1 - F.H. (STA. 1+44) and 1 - F.H. (STA 4+59) — real station-labeled callouts, Kingwood Drive sheet</i> 26 0813 - LSC Kingwood FTF_Aproved PUB Plans_260813.pdf p5 — <i>1 - F.H. (STA 4+18) — real station-labeled callout, Sorters Road sheet</i>
33	Gate valves with box, tees, and plugs/clamps at line junctions (12", 8", 6" sizes) Partially resolved by re-doing this properly instead of leaving it a flat guess. The STA 1+00 Kingwood Dr/Sorters Rd shared-boundary cluster IS confirmed as one real duplicate across sheets 04 and 05 (identical station, identical fitting composition — visually verified on the rendered sheets, not just text-matched) and is counted once, not twice, in any total. Every OTHER cluster on sheets 04-06 is now individually cited with its own station and composition (7 more distinct clusters: STA 1+44, STA 4+59, wet-connection end, STA 4+18, the Sorters Rd/crossing tie-in near STA 4+13, and two clusters on sheet 06 near the crossing itself). Left unresolved on purpose: one specific pair — the STA 4+13 tie-in cluster on sheet 05 ('1-12"x8" TEE, 1-8" G.V., 1-12" PLUG') and a nearby cluster on sheet 06 ('1-12" G.V., 1-12" PLUG') — sit close enough to the same physical junction that they might be one cluster split across two callout boxes, or two genuinely separate ones; their fitting compositions don't match closely enough to merge with confidence, and don't differ enough to separate with confidence either. That one junction needs a human with the full-size plan or the CAD file — not guessed at.	—	ea	unresolved	26 0813 - LSC Kingwood FTF_Aproved PUB Plans_260813.pdf p4 — <i>STA 1+00 junction cluster: 1-12"x12" TEE, 2-12" G.V. W/BOX, 1-12" PLUG & CLAMP</i> 26 0813 - LSC Kingwood FTF_Aproved PUB Plans_260813.pdf p5 — <i>Same STA 1+00 cluster repeated where sheet 05 shows the shared Kingwood Dr/Sorters Rd origin point — confirmed as the SAME physical cluster, not a second one, by matching both station and exact fitting composition</i>
33	8-inch fire-rated compound meter assembly (FRCWMA) with vault, in 31'x15' water meter easement	1	ea	cross_checked	26 0814 SOW Water Line 12 & 8 RV1.pdf p4 — <i>...8-inch fire-rated compound meter with vault (FRCWMA) in the 31'x15' water meter easement...</i> 26 0813 - LSC Kingwood FTF_Aproved PUB Plans_260813.pdf p6 — <i>PROP. FRCWMA W/ VAULT, labeled on the Sorters Road Crossing plan and profile</i>

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33	8-inch reduced pressure zone (RPZ) backflow preventer assembly, on private property	1	ea	cross_checked	26 0814 SOW Water Line 12 & 8 RV1.pdf p4 — ...and the 8-inch reduced pressure zone (RPZ) backflow preventer assembly located on private property... 26 0813 - LSC Kingwood FTF_Aproved PUB Plans_260813.pdf p6 — PROPOSED 8" REDUCED PRESSURE ZONE (RPZ) BACKFLOW PREVENTOR IN PRIVATE PROPERTY — key note W2, Sorters Road Crossing sheet
33	System design capacity	272	gpd	schedule_verified	26 0818 SOW Septic System.pdf p1 — The system shall be constructed to accommodate a maximum daily usage of 272 Gallons Per Day (GPD).
33	Minimum required trench bottom area	1,360	sf	schedule_verified	26 0818 SOW Septic System.pdf p1 — The contractor shall prepare a minimum required trench bottom of 1,360 square feet...
33	Engineered drip disposal area	2,560	sf	schedule_verified	26 0818 SOW Septic System.pdf p1 — ...and install the engineered drip area across 2,560 square feet.
33	Pretreatment tank, 500-gallon	1	ea	schedule_verified	26 0818 SOW Septic System.pdf p1 — Furnish, excavate, and install a 500-gallon pretreatment tank...
33	Clearstream aerobic treatment unit, 800-gallon	1	ea	schedule_verified	26 0818 SOW Septic System.pdf p1 — ...an 800-gallon Clearstream Treatment Unit...
33	Pump tank, 1,080-gallon SOW explicitly states no chlorinator is required — do not price one.	1	ea	schedule_verified	26 0818 SOW Septic System.pdf p1 — ...and a 1,080-gallon pump tank. A chlorinator is not required for this specific design.
33	Simplex effluent pump, Goulds 20 EB, 0.5 HP	1	ea	schedule_verified	26 0818 SOW Septic System.pdf p1 — ...a Simplex Effluent Pump (Goulds 20 EB 1/2 HP)...
33	Solids pump, 2-inch, Goulds, in dosing tank	1	ea	schedule_verified	26 0818 SOW Septic System.pdf p1 — ...a 2-inch Goulds Solids Pump inside the dosing tank...
33	Effluent pump, Goulds, 0.5 HP, 20 GPM	1	ea	schedule_verified	26 0818 SOW Septic System.pdf p1 — ...and a 0.5 HP Goulds 20 GPM Effluent Pump.
33	Dual filter, Aztec, 100 micron	1	ea	schedule_verified	26 0818 SOW Septic System.pdf p2 — Install an Aztec Dual 100 Micron Filter to ensure effluent clarity and protect the drip infrastructure.
33	Drip tubing, Netafim, 0.91 GPH, across the 2,560 SF drip field Quantity re-stated from the drip-area line above — this is the area the tubing must cover, not a separate linear-footage figure (none given; drip emitter spacing/lateral spacing needed to convert to LF of tubing is not in the SOW).	2,560	sf	schedule_verified	26 0818 SOW Septic System.pdf p2 — Install Netafim 0.91 GPH drip tubing throughout the designated absorption area.
33	Supply piping, training facility building to septic system (Sch 40 or DWV) No length given in the SOW; the septic system's own 2-page SOW has no site plan showing the building-to-tank distance. Needs the actual site/civil plan for the fire training facility building, which is not included in this bid-attachments package.	—	lf	unresolved	26 0818 SOW Septic System.pdf p1 — Install Schedule 40 or DWV thick-wall pipe for all connections running from the training facility building to the septic system.